

Business Flow — Fathom Maritime Operations & Compliance

This document captures the end-to-end business journey of the system, the roles involved, the lifecycle of each entity, and how compliance is computed.

1. Roles

Admin (Fleet / Operations Manager)

- Owns the fleet and crew records.
- Plans maintenance work and schedules safety drills.
- Monitors compliance across all ships.

Crew (Officers / Engineers / Bosun / etc.)

- Belongs to one ship.
- Receives the maintenance tasks assigned to them.
- Sees the upcoming drills for their ship and records their own attendance.
- Logs notes / status updates on their tasks.

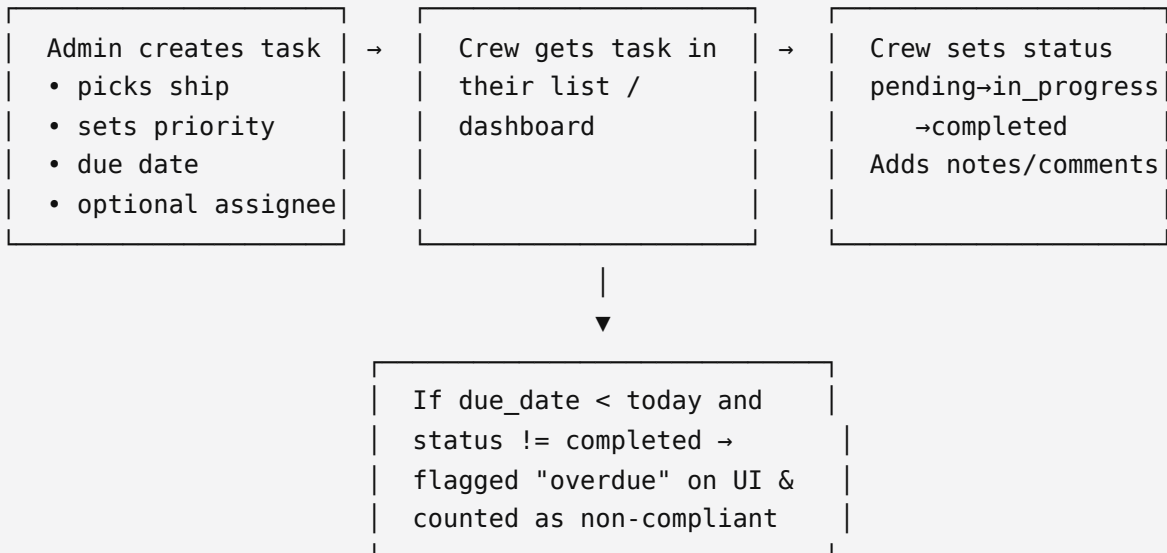
2. Core entities

Entity	Purpose
Ship	A vessel in the fleet, identified by name + IMO number.
User	Admin or crew member; crew are anchored to a ship.
MaintenanceTask	Work item against a ship, with a due date and priority.
TaskComment	Notes posted on a maintenance task by admin or assignee.
Drill	A scheduled safety exercise (fire, evacuation, etc.) on a ship.
DrillParticipation	Per-crew-member attendance record for a given drill.



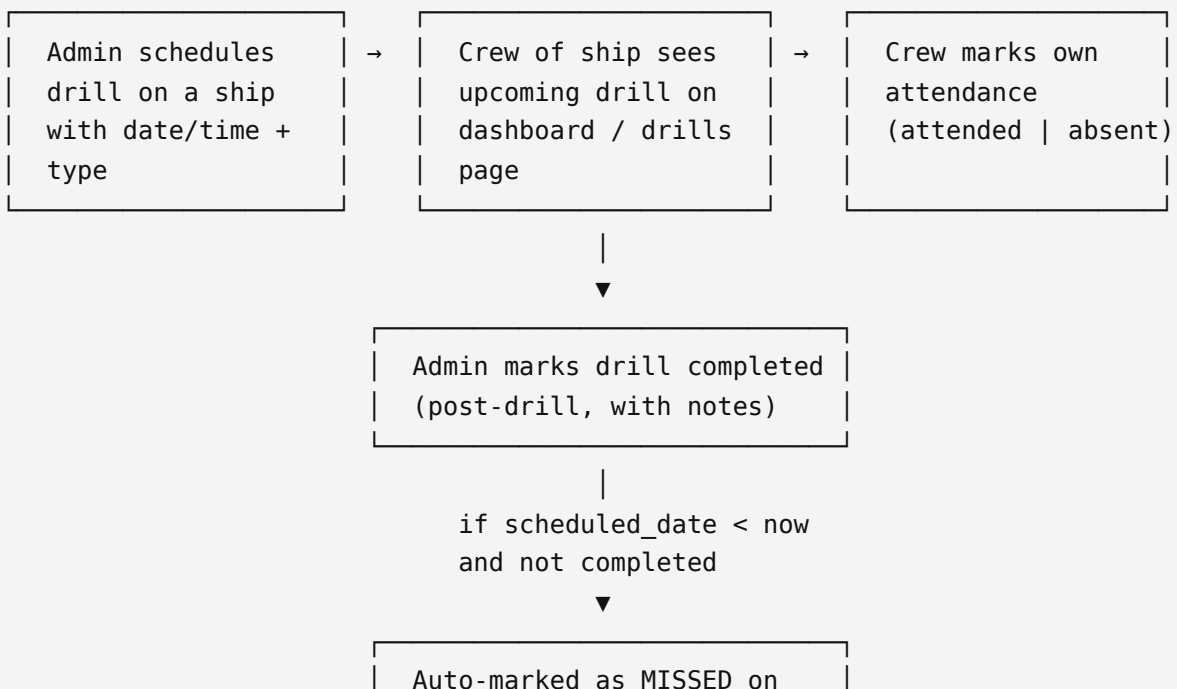
Drill *——* DrillParticipation *——* User (crew)

3. Maintenance flow



Crew can only see/update tasks assigned to them; the API enforces this on every read and write.

4. Drill flow



next compliance read

`autoMarkMissedDrills` is invoked at the start of every compliance dashboard load. This avoids needing a separate cron and keeps the data self-healing even on a fresh deployment.

5. Compliance computation

Inputs (per ship)

- `T_total` = number of maintenance tasks ever created
- `T_done` = tasks with status = completed
- `T_overdue` = tasks past due date and not completed
- `D_elapsed` = drills whose scheduled date is in the past
- `D_done` = drills with status = completed (within `D_elapsed`)
- `D_missed` = drills past scheduled date but not completed
- `attendance_marks` = sum across `D_done` of crew members marked `attended`
- `crew_size` = active crew users assigned to this ship

Per-ship metrics

```
maintenance_pct      = (T_done / T_total)      × 100
drill_completion_pct  = (D_done / D_elapsed)    × 100    (100 if no elapsed)
participation_pct     = (attendance_marks /
                        (crew_size × D_done))    × 100    (100 if no D_done)

overall_score = round(
    maintenance_pct      × 0.50
  + drill_completion_pct × 0.30
  + participation_pct    × 0.20
)
```

Classification

Score range	Class
≥ 90	<code>compliant</code>
70 – 89	<code>at_risk</code>
< 70	<code>non_compliant</code>

Fleet-level overall is the simple average across all ships' overall scores.

Why this weighting?

Maintenance is the largest carrier of operational risk on a ship, so it carries the highest weight (0.5). Drill completion vs participation is weighted 0.3 + 0.2 because completing the drill is more visible at the fleet level than per-crew participation, but both still matter to the regulator's view of safety culture.

6. Risk highlighting

The compliance API for each ship returns explicit lists of:

- `overdue_tasks` — id, title, due date, priority, status
- `missed_drills` — id, title, scheduled date, drill_type

The admin dashboard also highlights ships with `non_compliant` / `at_risk` classifications and shows count of overdue / missed items.

Crew see overdue tasks highlighted in their personal list (rose-tinted row, red due date) so they can self-prioritise.

7. Auth & access control

- Login at `/api/auth/login` returns a JWT with `{ id, role }` payload.
 - Every subsequent call must send `Authorization: Bearer <token>`.
 - `requireAuth` middleware loads the fresh user record from DB on every call (so deactivating a user revokes them within one request).
 - `requireRole('admin')` is layered on top of routes that change schema-level data (creating ships, crew, tasks, drills).
 - Crew's listing endpoints automatically scope to:
 - tasks: `assigned_to = me`
 - drills: `ship_id = my ship`
 - task detail: must be the assignee
 - drill detail: must belong to my ship
-

8. Sample storyboard

1. **Admin** logs in → fleet dashboard shows score = 43% (non-compliant) with 3 overdue tasks and 2 missed drills.
2. Admin clicks *Maintenance* → filters by ship `MV Aurora` and `overdue` → sees `Lifeboat hoist inspection` flagged red. Reassigns it to the master.
3. Admin clicks *Drills* → schedules a `man_overboard` drill for `MV Aurora` four days out.

4. **Captain Rao** (crew on Aurora) signs in → his dashboard shows the newly-assigned task and the upcoming drill.
5. Captain opens the task → posts a note "Spares ETA Friday", sets status to `in_progress` . The dashboard score recomputes on next load.
6. On the day of the drill, Captain visits the drill and marks himself `attended` with remarks "All hands present".
7. Admin opens the drill, adds notes "Smooth, 7 minutes", marks it complete.
8. Fleet dashboard refreshes — `MV Aurora` 's drill completion + participation percentages climb, overall score nudges upward.